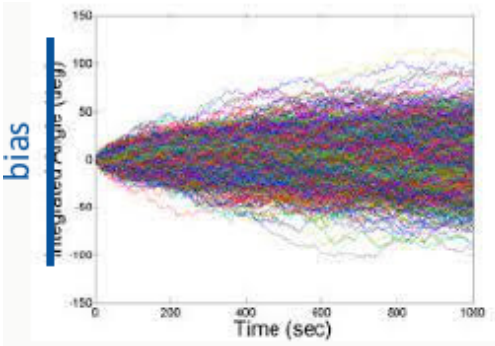
Ms08: GNSS/IMU Integration

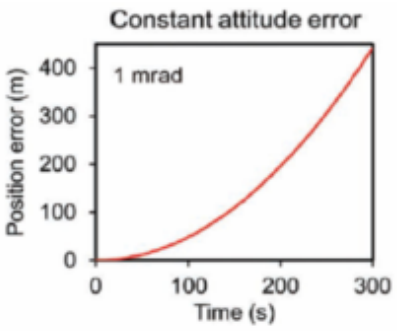
Inertial Navigation Systems

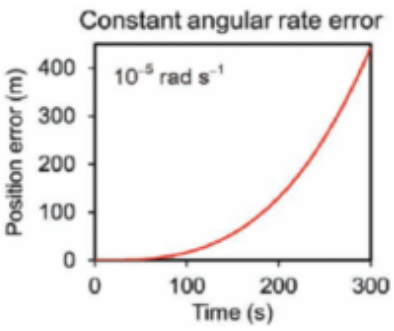
- 3 axis gyro
- 3 axis accelerometer
- GNSS receiver with 1 or 2 antennas
- **Goal**:
  - 3D position
  - 3D Orientation
  - 3D Velocity

GNSS/IMU Integration

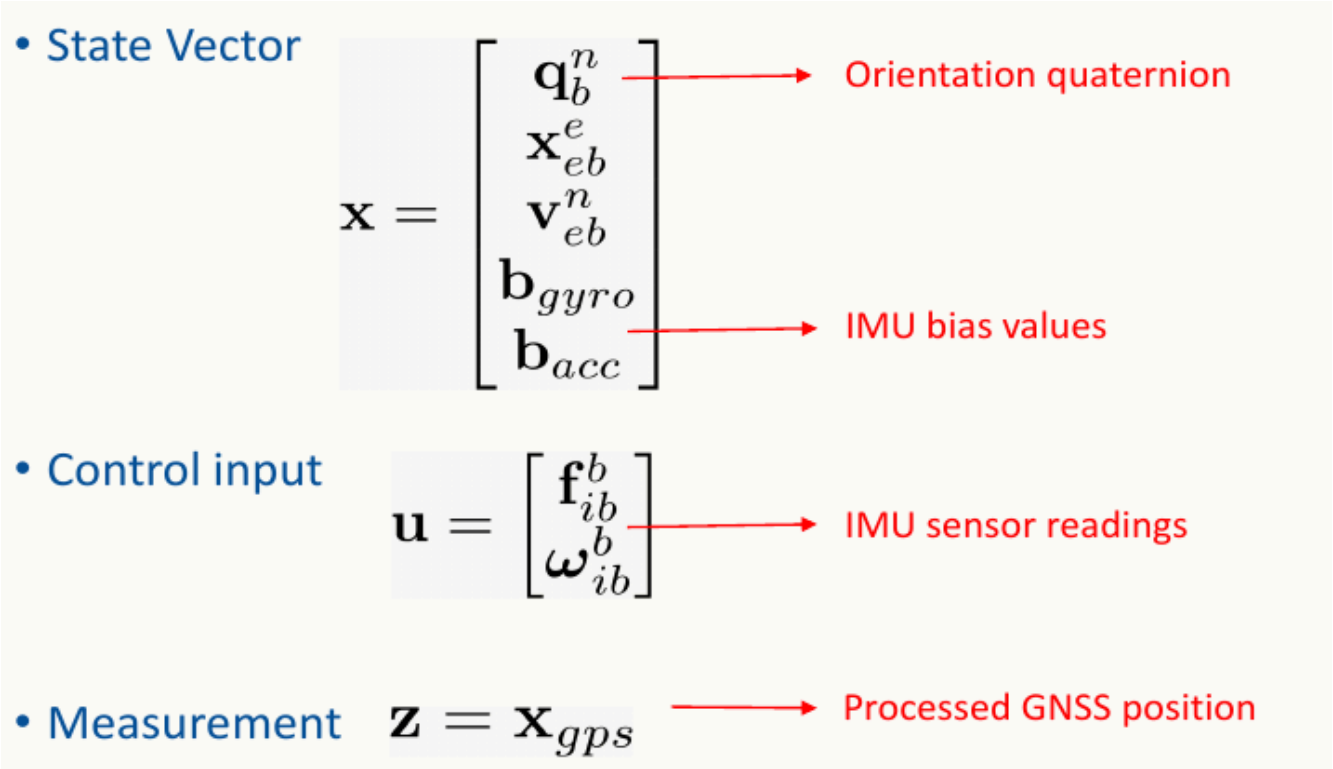
- Estimate position & orientation of the system in all dimensions
- Use inertial sensors (acc + gyro) as control input:
  - Strapdown inertial integration as system model
- Error of acc + gyro: Time **dependent systematic errors** (bias)
  - Estimate biases as part of the state vector
- Use GNSS in measurement model

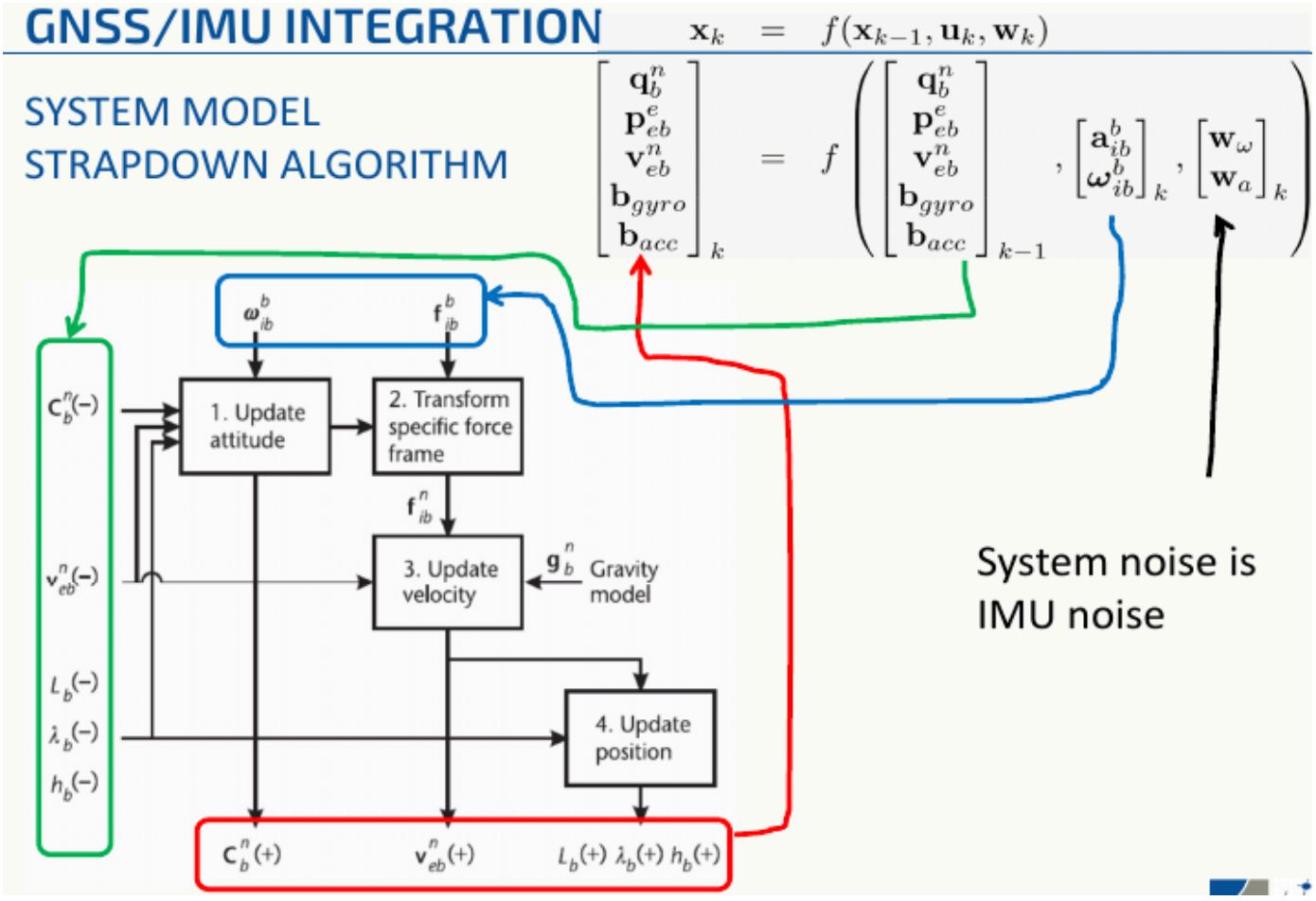






- State Vector

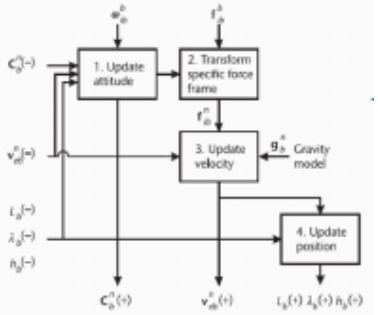




GPS/IMU INTEGRATION

MEASUREMENT MODEL

$$\mathbf{x} = \begin{bmatrix} \mathbf{q}_b^n \\ \mathbf{x}_{eb}^e \\ \mathbf{v}_{eb}^n \\ \mathbf{b}_{gyro} \\ \mathbf{b}_{acc} \end{bmatrix}$$



$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

- Measurement is only GPS Position $x_{gps}, y_{gps}, z_{gps}$

$$\mathbf{z}_k = \begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ z_{gps,k} \end{bmatrix} = h(\mathbf{x}_k) = \mathbf{H}\mathbf{x}_k = \begin{bmatrix} \dots & 1 & 0 & 0 & \dots \\ \dots & 0 & 1 & 0 & \dots \\ \dots & 0 & 0 & 1 & \dots \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 & 0 \\ 0 & \sigma_{gps,y}^2 & 0 \\ 0 & 0 & \sigma_{gps,z}^2 \end{bmatrix}$$

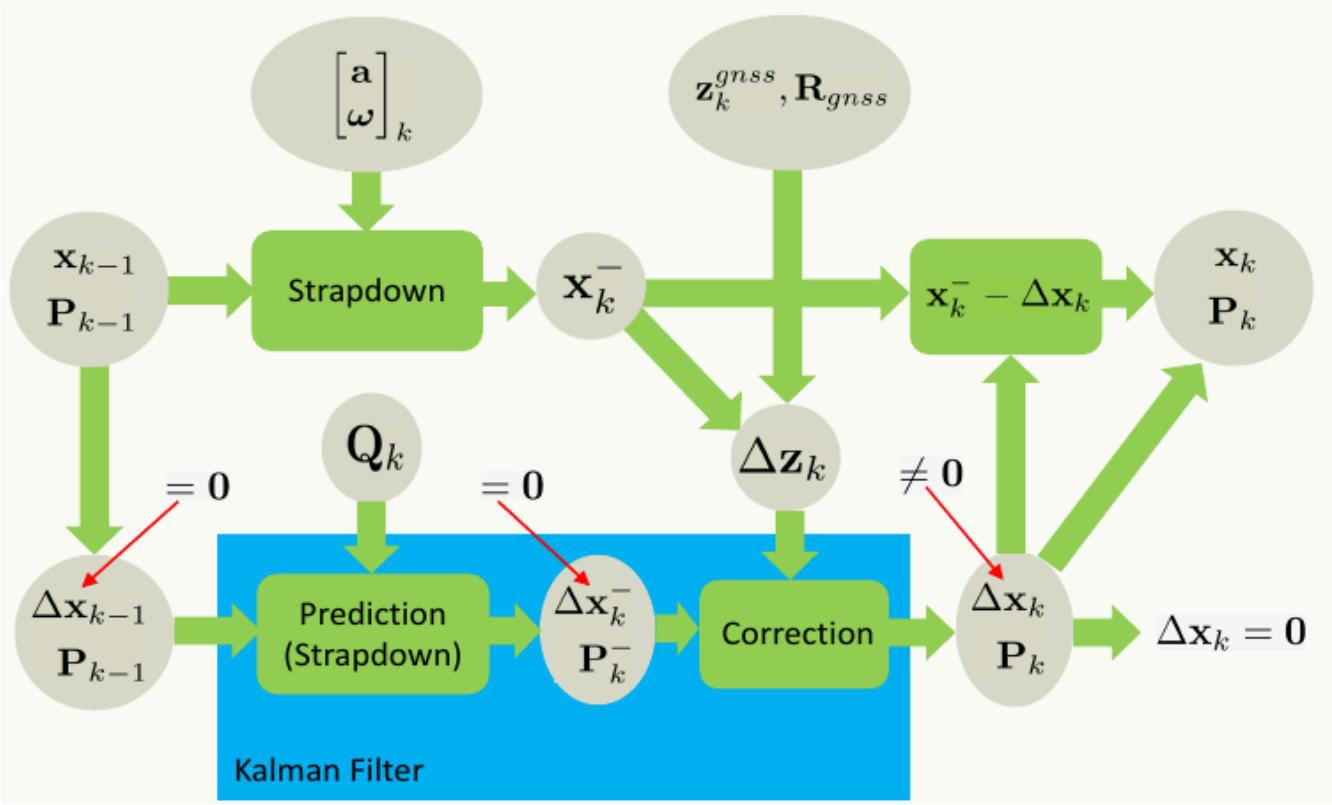
- But Remember: GPS noise is not white
  - R is wrong

Tightly Coupled Vs. Loosely Coupled

- Using a **processed GPS position** in the measurement model:
  - **Loosely Couple Integration**
- Using **raw** GNSS observation (carrier phases, pseudo ranges):
  - **Tightly Coupled Integration**
  - GNSS processing (+ ambiguity fixing) within filter
  - Better ambiguity fixing, because IMU data Integration
  - Included Cycle slip detection
  - Observation from a small number of satellite (<4) can be integrated into filter
  - More complex filter algorithm
- Comparable results for high satellite visibility

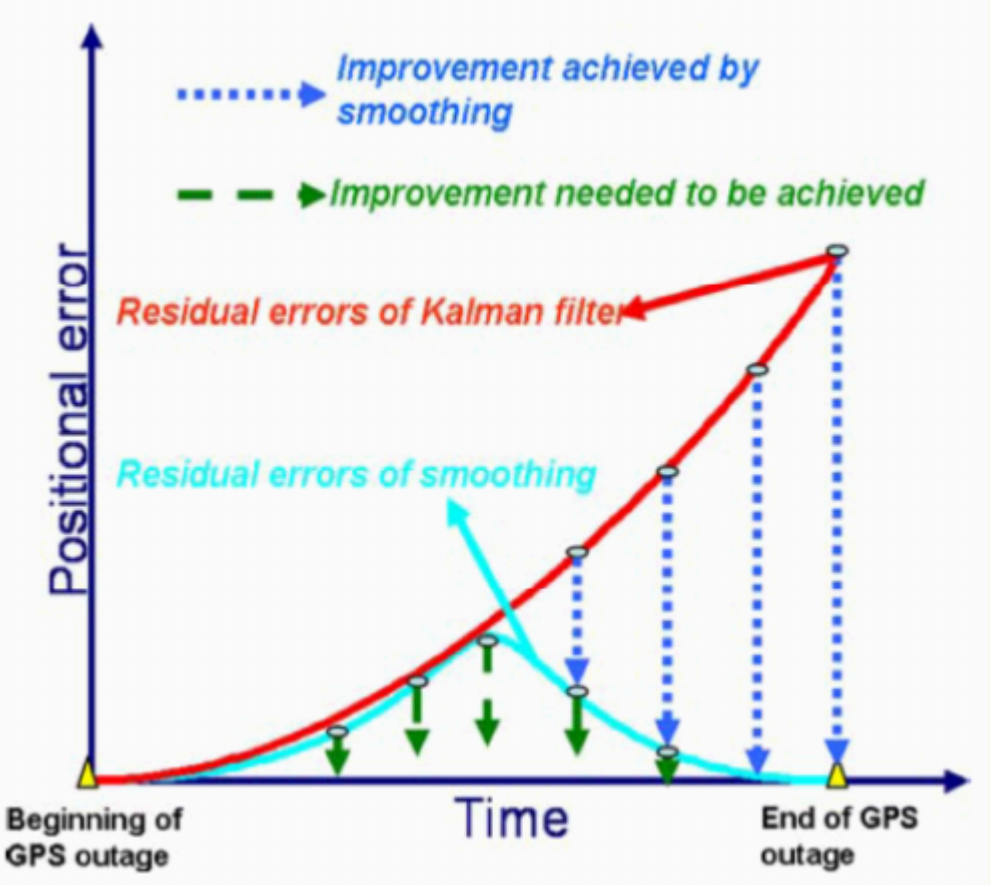
Error State Filtering

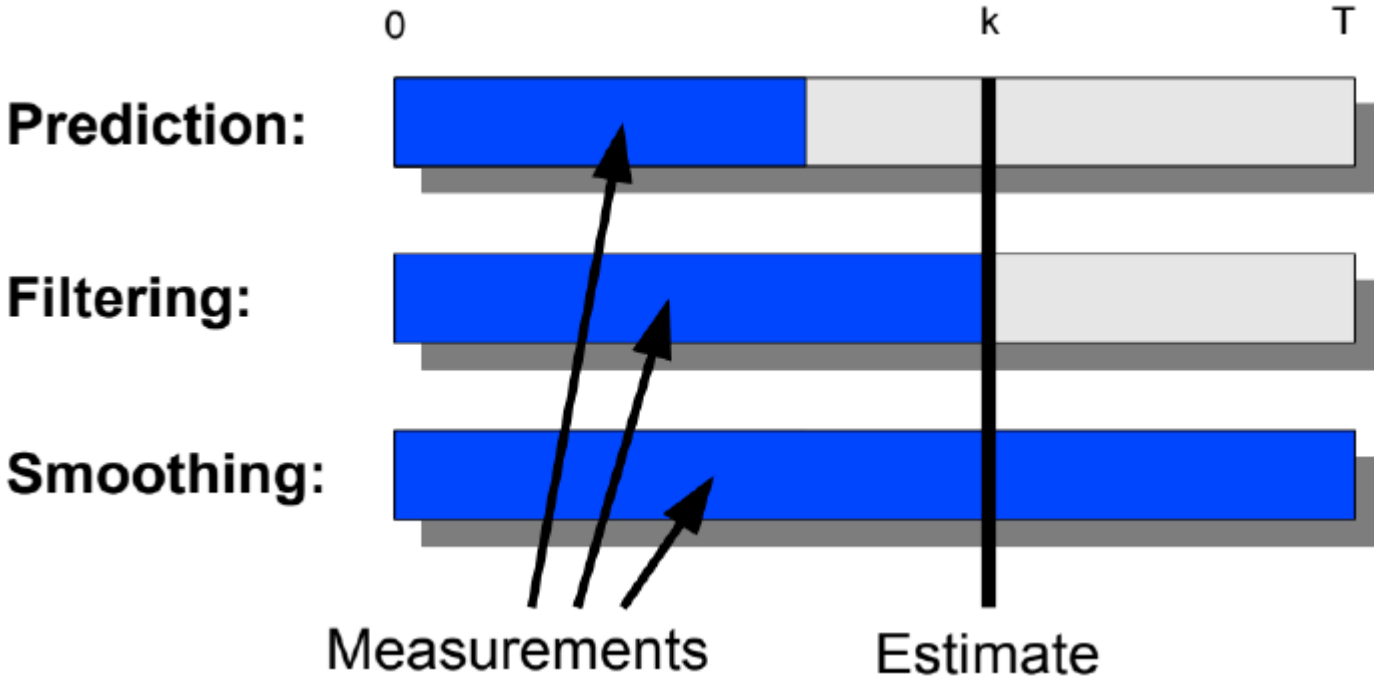
- GNSS/IMU Fusion Error State KF (ESKF)



- Error State Filtering
  - Theoretically identical
  - Potentially more numerically stable
  - Common realisation of GNSS/IMU Fusion Kalman Filter
  - Error dynamics are lower than motion dynamics
    - Linearisation is not that critical
    - Error Filter update rate can be smaller

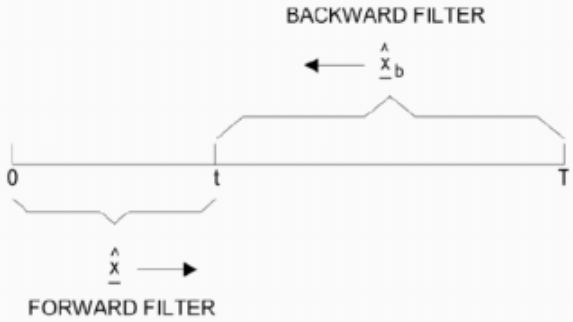
Kalman Smoothing





Forward-Backward Filter

FORWARD-BACKWARD FILTER

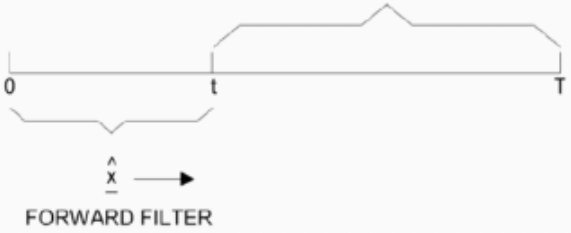


Weighted mean of forward and backward filter

$$
\begin{aligned}
\mathbf{x}_{S,k} &= K_{F,k}\mathbf{x}_{F,k} + K_{B,k}\mathbf{x}^{-}_{B,k} \\
&= K_{F,k}\mathbf{x}_{F,k} + (1 - K_{F,k})\mathbf{x}^{-}_{B,k} \\
K_{F,k} &= P^{-}_{R,k}(P_{F,k} + P^{-}_{B,k})^{-1} \\
P_{S,k} &= \left[(P_{F,k})^{-1} + (P^{-}_{B,k})^{-1}\right]^{-1}
\end{aligned}
$$

RTS Smoother

1) Start with a forward filter from 0 to T
2) Run a „smoother“ from T to 0

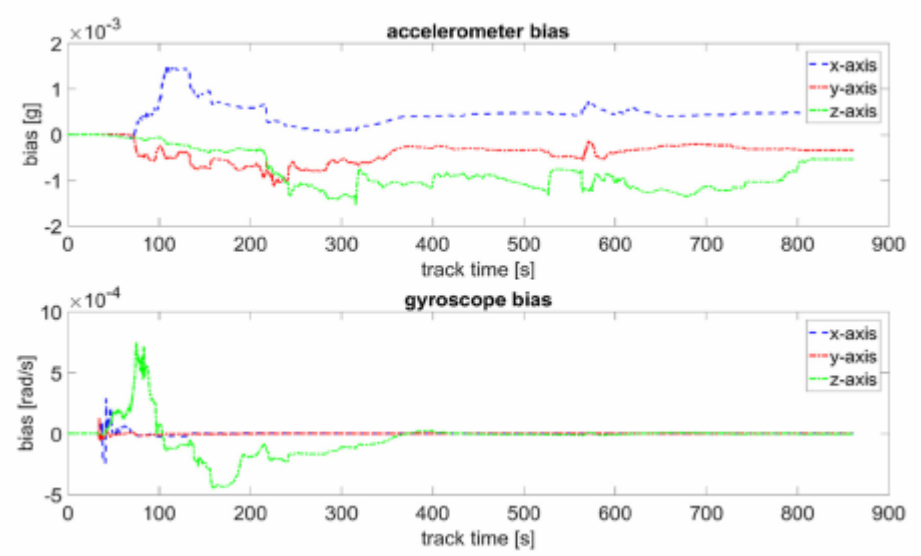


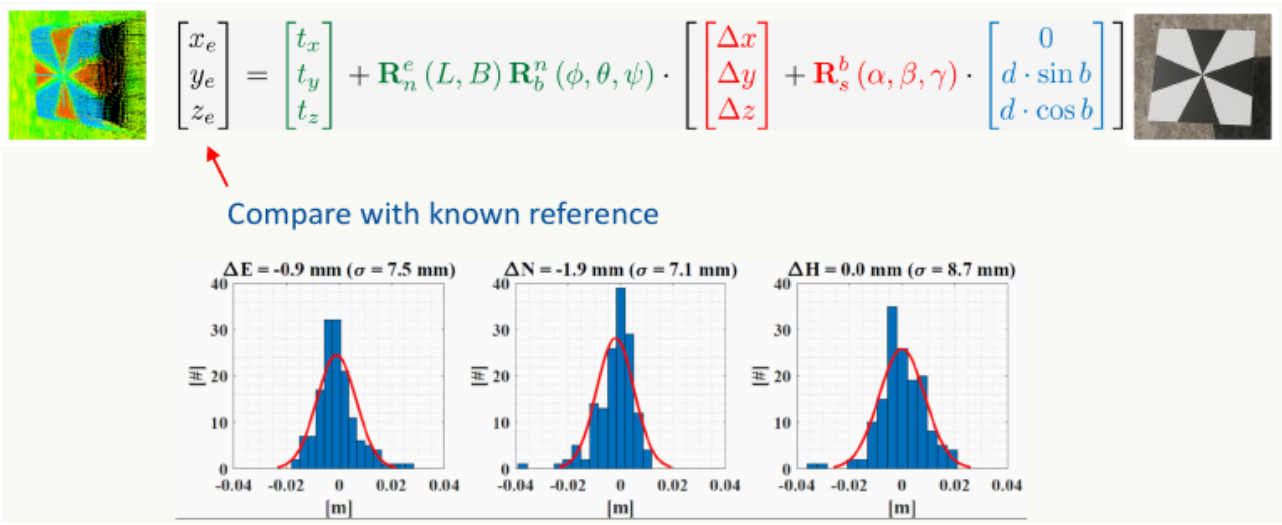
$$
\begin{aligned}
\mathbf{x}_{S,k} &= \mathbf{x}_{F,k} + A_k(\mathbf{x}_{S,k+1} - \mathbf{x}^{-}_{F,k+1}) \\
A_k &= P_{F,k}\Phi_k^T\left[P^{-}_{F,k+1}\right]^{-1} \\
P_{S,k} &= P_{F,k} + A_k(P_{S,k+1} - P^{-}_{F,k+1})
\end{aligned}
$$

More efficient. Only need to store forward states and covariances

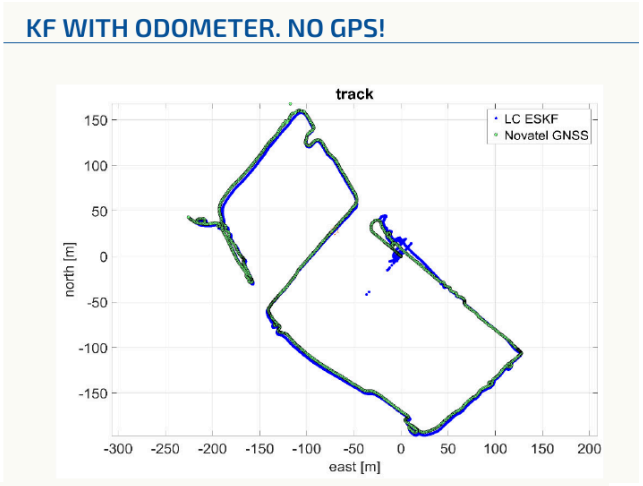
Trajectory Estimation / Quality Evaluation

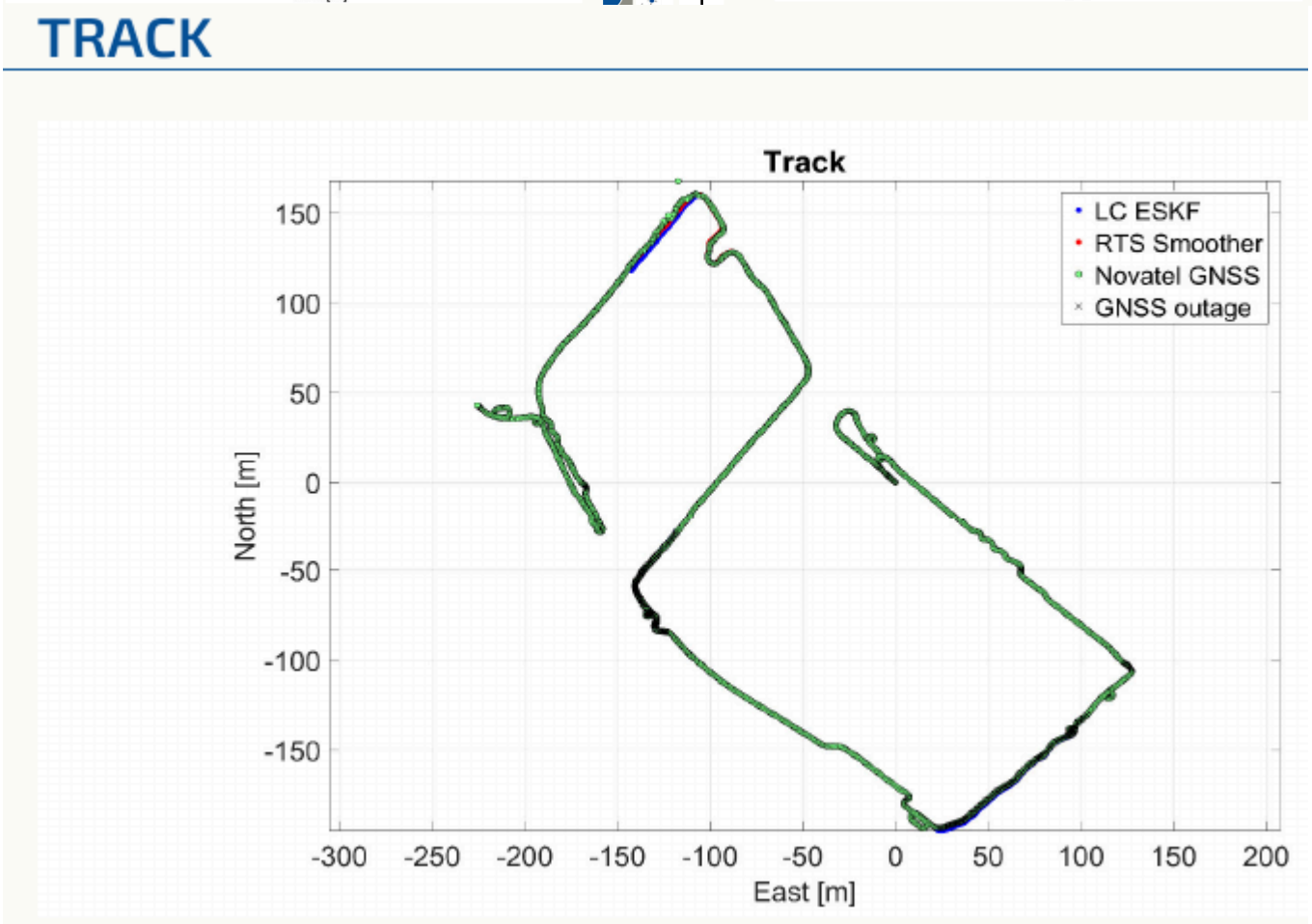
- Look at **Covariance** of the Estimate P_k
  - Does **not depend on measurements**
  - Only valid, if all models and uncertainties are **correct**
  - **No systematic errors** included

- Look at **Innovation** $(z_k - H\boldsymbol{x}_k^-)$
  - Gives idea on consistency between models and observations
  - Not clear if inconsistencies come from **wrong measurement** or **predictions**
- Look at Estimated States
  - Nice looking position result can also appear if model completely wrong and measurement are good
  - E.g. Estimated bias values show consistency of models
  - 

- Compare with reference Solution
  - Needs synchronisation between reference and estimation results
  - Difficult to realise in outdoor scenarios (especially orientation)
  - Error depends on motion, environment, time -> impossible parametrisation
- Indirect Evaluation
  - Includes many other errors
  - Assignment of error to specific sources is difficult
  - 

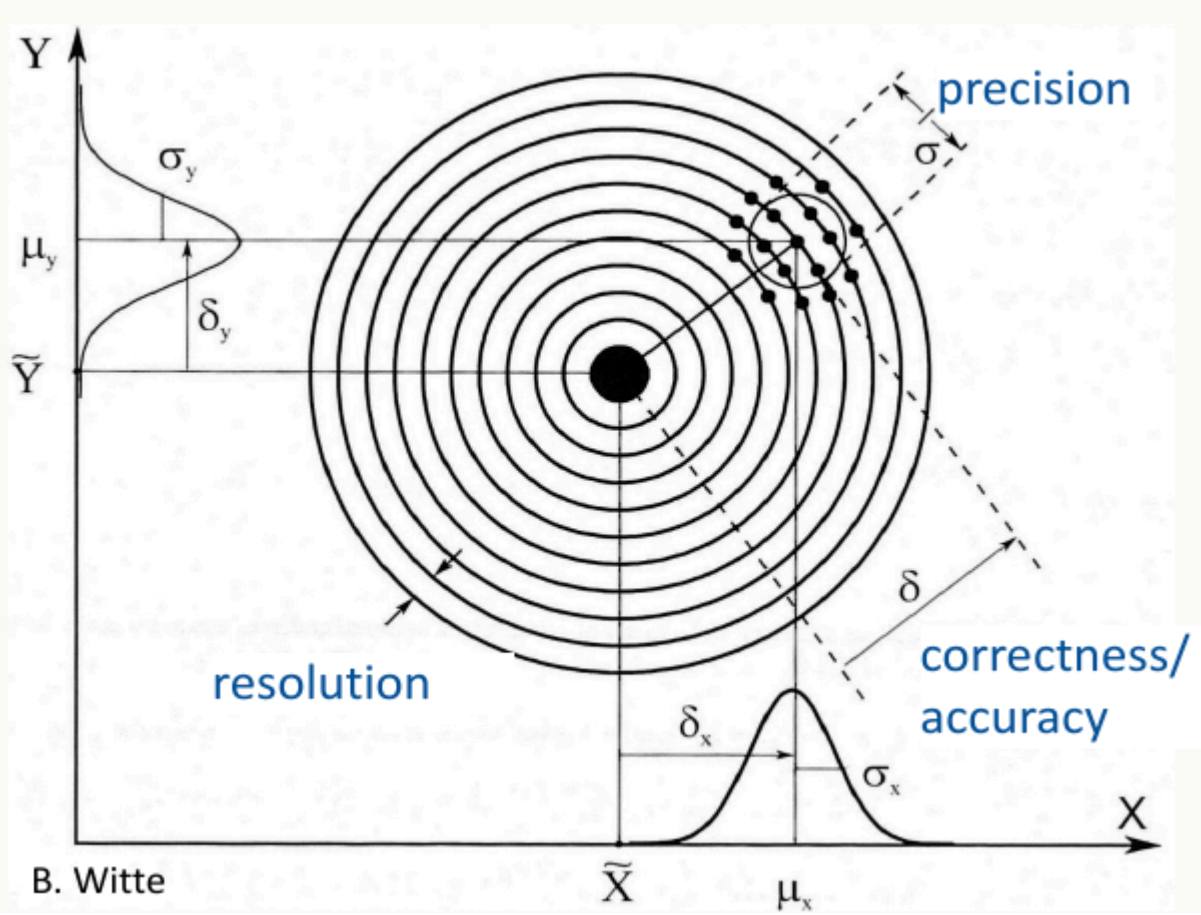
Evaluation without Reference Information



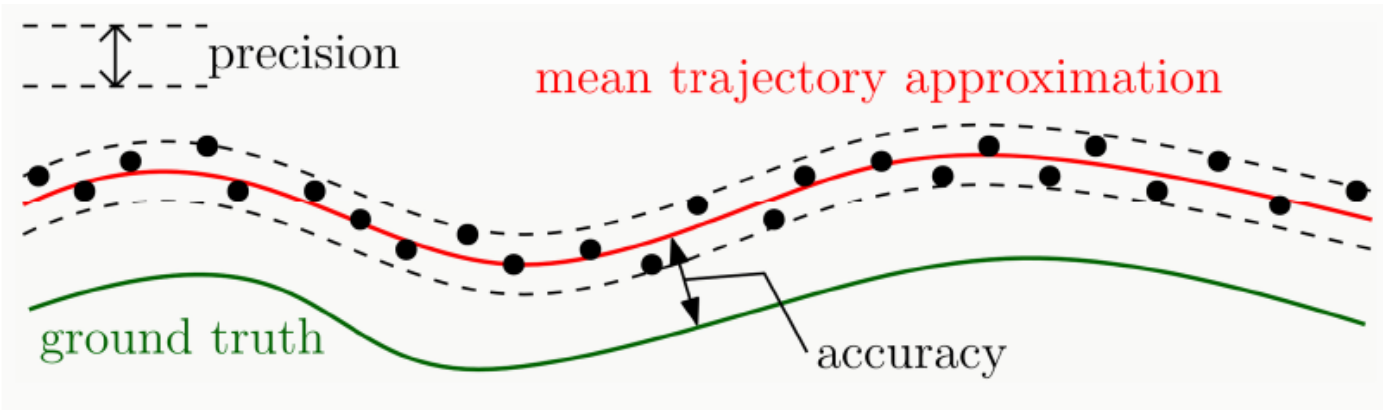


- **Track comparison**: RTS smoother without and with GNSS outages
- **GNSS Quality**
  - Dependent on Number of GNSS satellites
- Difference to GNSS data
- Bias estimation

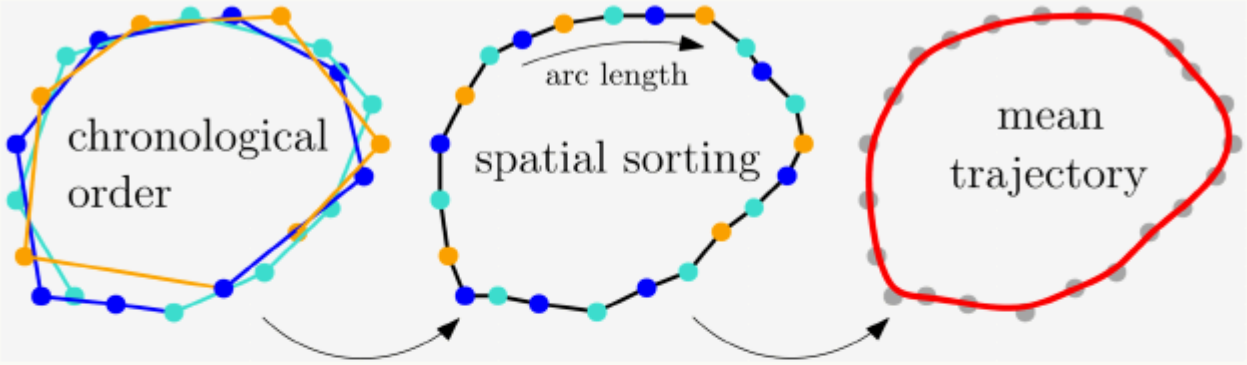
Trajectory Evaluation using Repeated Measurements on a Reference Track

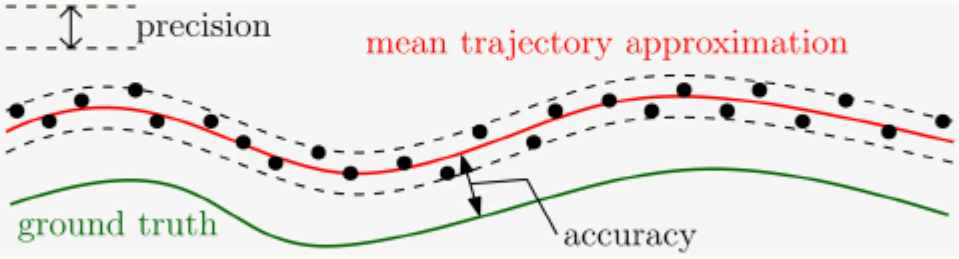


- Point-to-point comparison need a synchronised reference with higher accuracy
- Precision evaluation need repeated measurements and the possibility to build a mean value



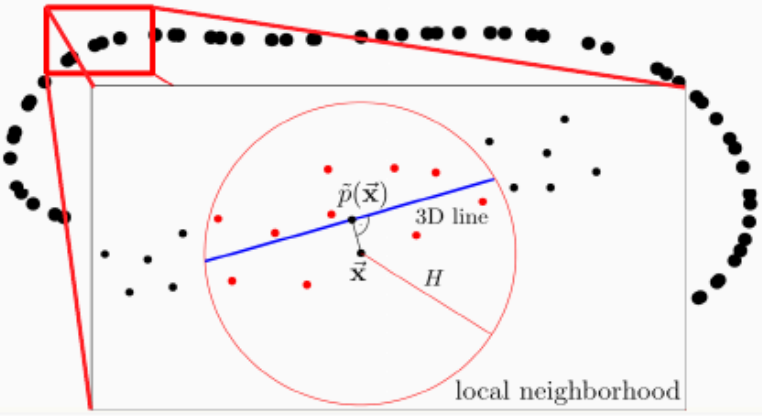
EVALUATION USING REPEATED MEASUREMENTS



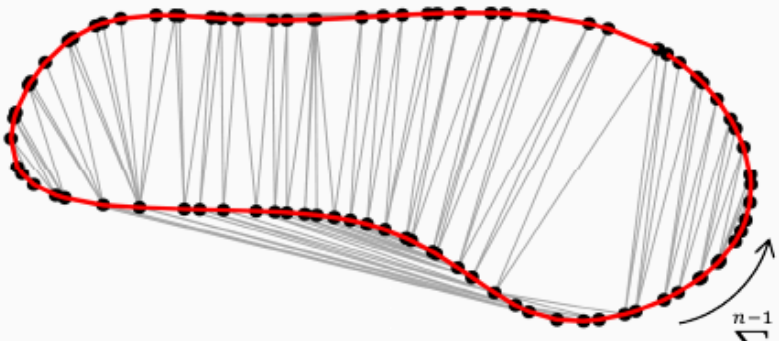


1. spatial sorting
2. approximation
3. quality parameters

*STEP 1: SPATIAL SORTIERUNG



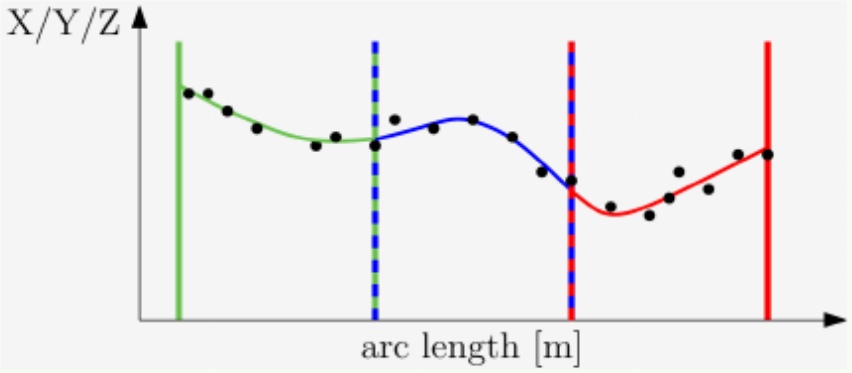
- **Smoothing** using Moving-Least-Squares Approximation



- **Sorting** using Minimum-Spanning-Tree (sorting parameter: arc length)

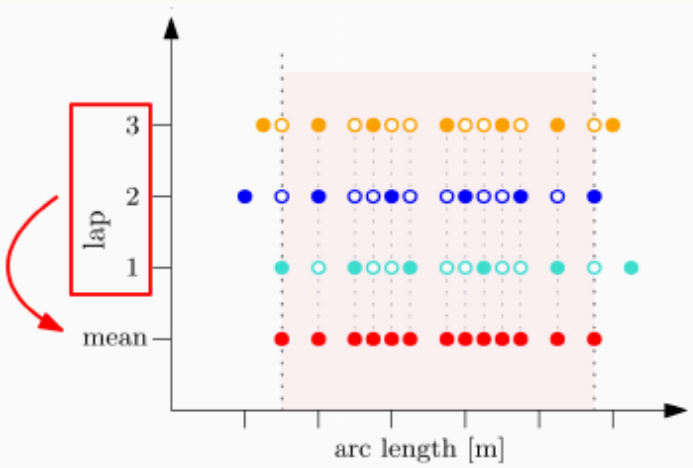
arc length $l_{arc} = \sum_{i=1}^{n-1} \|p_{i+1} - p_i\|$

*STEP 2: APPROXIMATION



Positions:

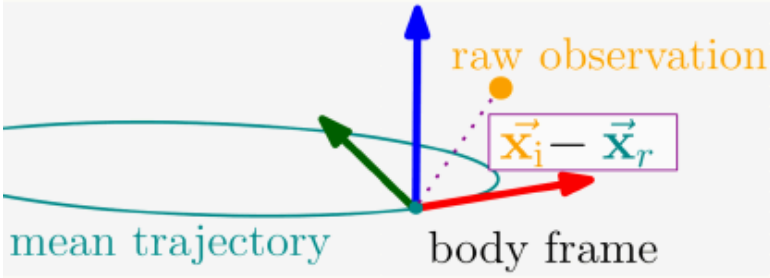
Cubic spline approximation



Orientations:

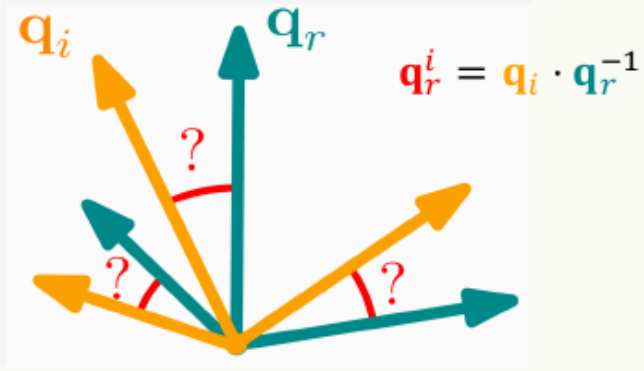
Interpolation and averaging

*STEP 3: PARAMETER EXTRACTION



Positions:
Deviations relative to body frame

➔ Cross track deviations



Orientation:
Differences of previously interpolated rotations

➔ roll, pich, yaw